

Context

In this activity, we will show you how to install PT and GNS3, which, like “packet tracer,” is a CISCO network simulator. Installing PT should not be problematic regardless of the machine (PC) used. When implementing GNS3, we will illustrate the installation with a virtual machine (VM) using a web browser (Google Chrome), as this is how it should run on any PC, even if it is not high-performance.

1 – Install Cisco Packet Tracer

You need to create an account and access the following URL:

<https://skillsforall.com/resources/lab-downloads>

Complete the activities “Getting Started with Cisco Packet Tracer” and “Exploring Networking with Cisco Packet Tracer” (optional). These are guided activities where you can familiarize yourself with the simulator and understand its interactivity as well as the basic principles of network configuration.

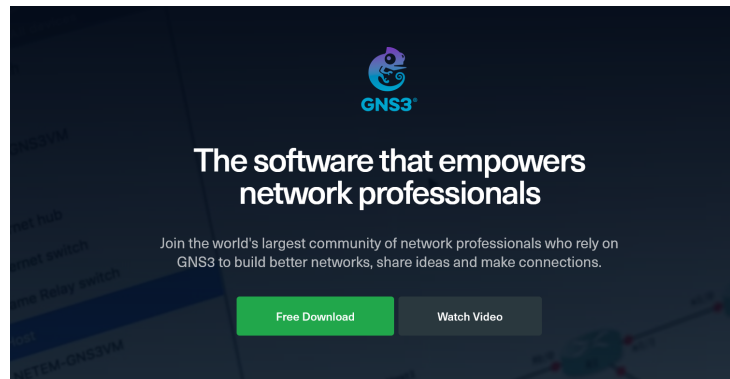
2 – Installing GNS3

The steps recommended here can also be viewed in the following video: link.

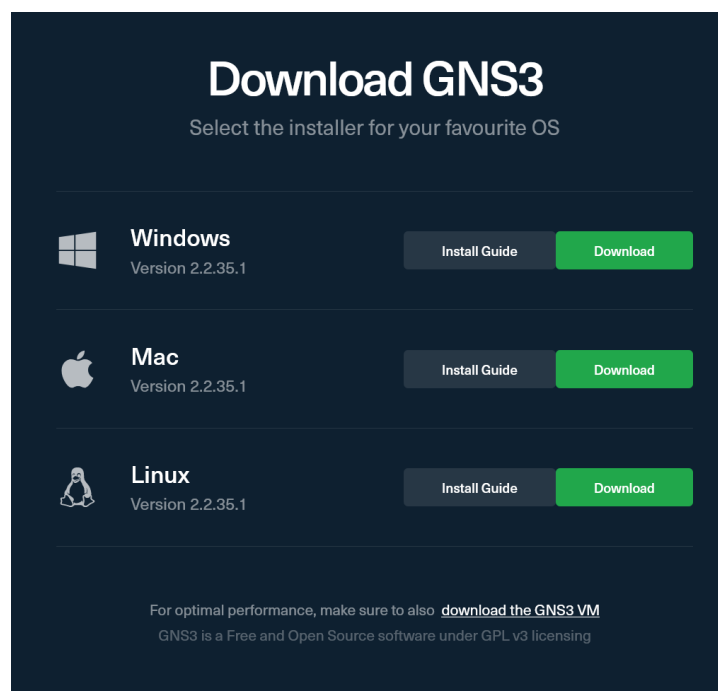
1. Install VMware software (free license for non-commercial use)

<https://www.vmware.com/go/getplayer-win>

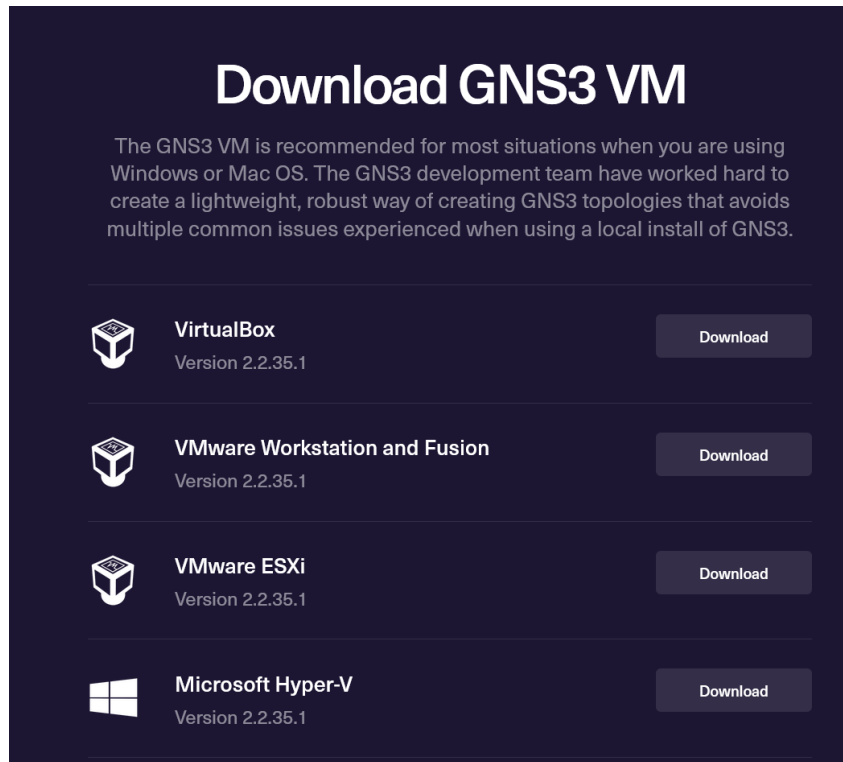
2. Install GNS3. To ensure stability, we recommend the version that can be found at the following link. The latest version can be found at <https://www.gns3.com/>:



It is recommended to install GNS3 VM.



The VMware Workstation option is recommended



Configure the GNS3 VM

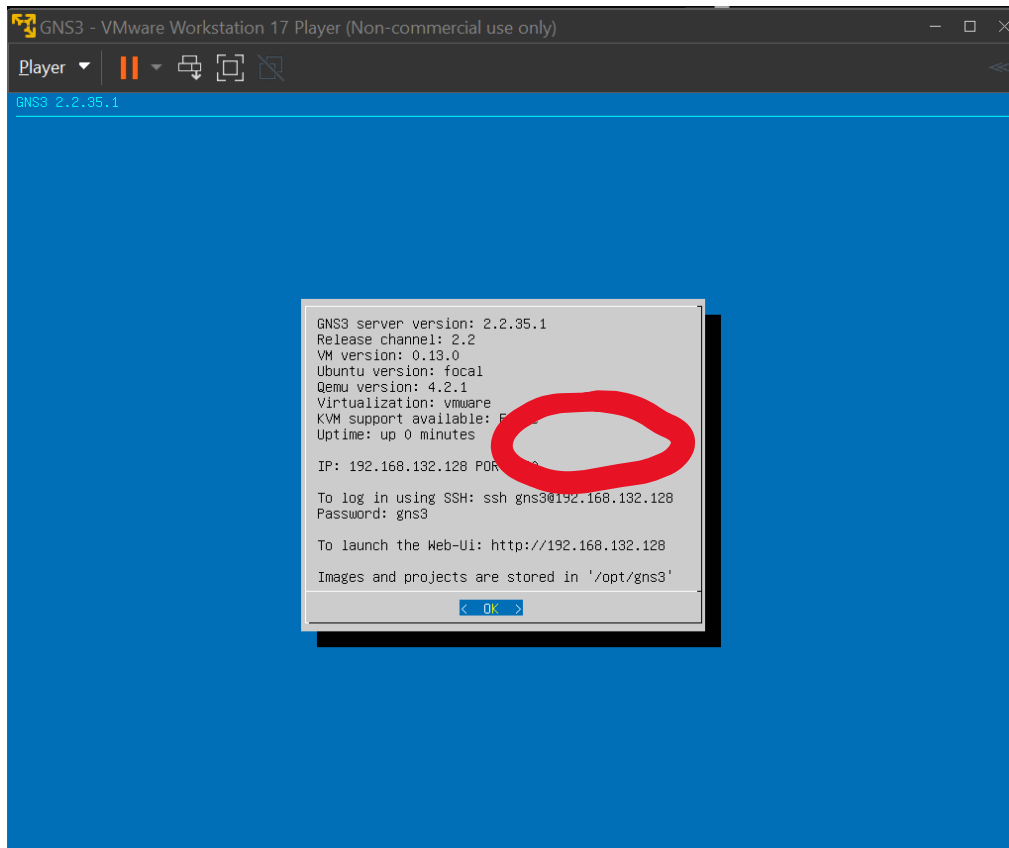
After the download is complete, a file with the .ova extension is created in the folder selected for the destination.

☐ Nome	Estado	Data de modificação	Tipo	Tamanho
 GNS3 VM		18/11/2022 09:15	Ficheiro OVA	968 436 KB

This file must be executed and the instructions followed to define the configuration details of the GNS3 VM, namely its name and path.

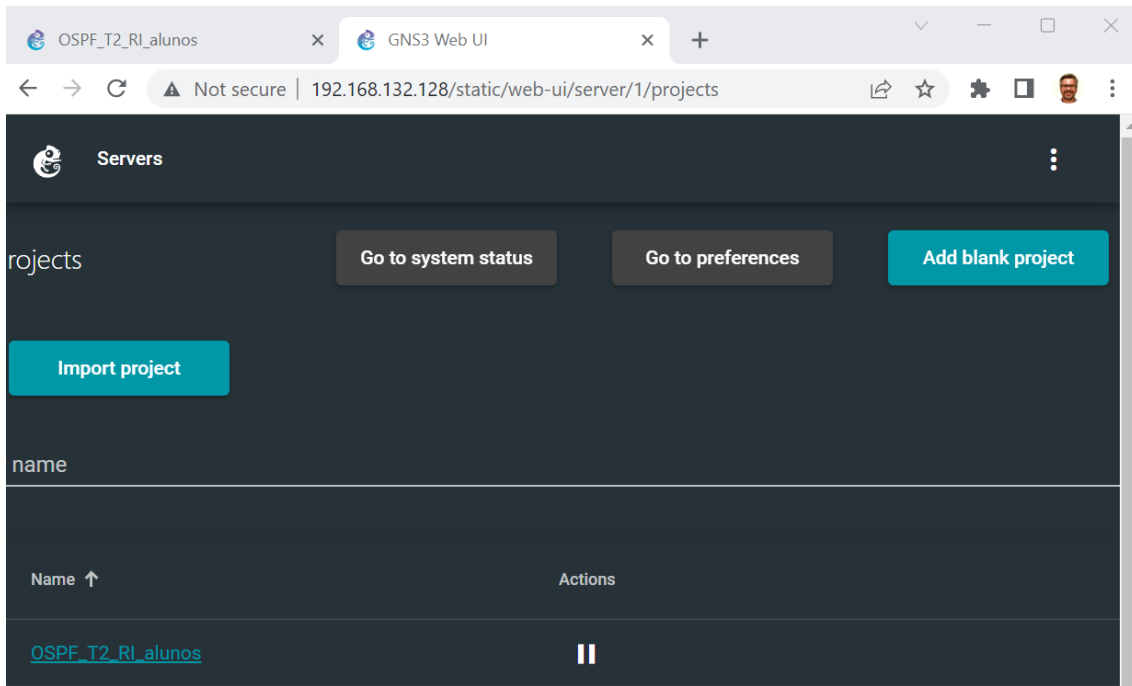
Start the VM

Run the software already installed in VMware Workstation Player 17 and click the “play” button to load the VM, as defined in the previous step. Once loading is complete, the IP address will appear, which can be copied to the browser to access the Web UI.

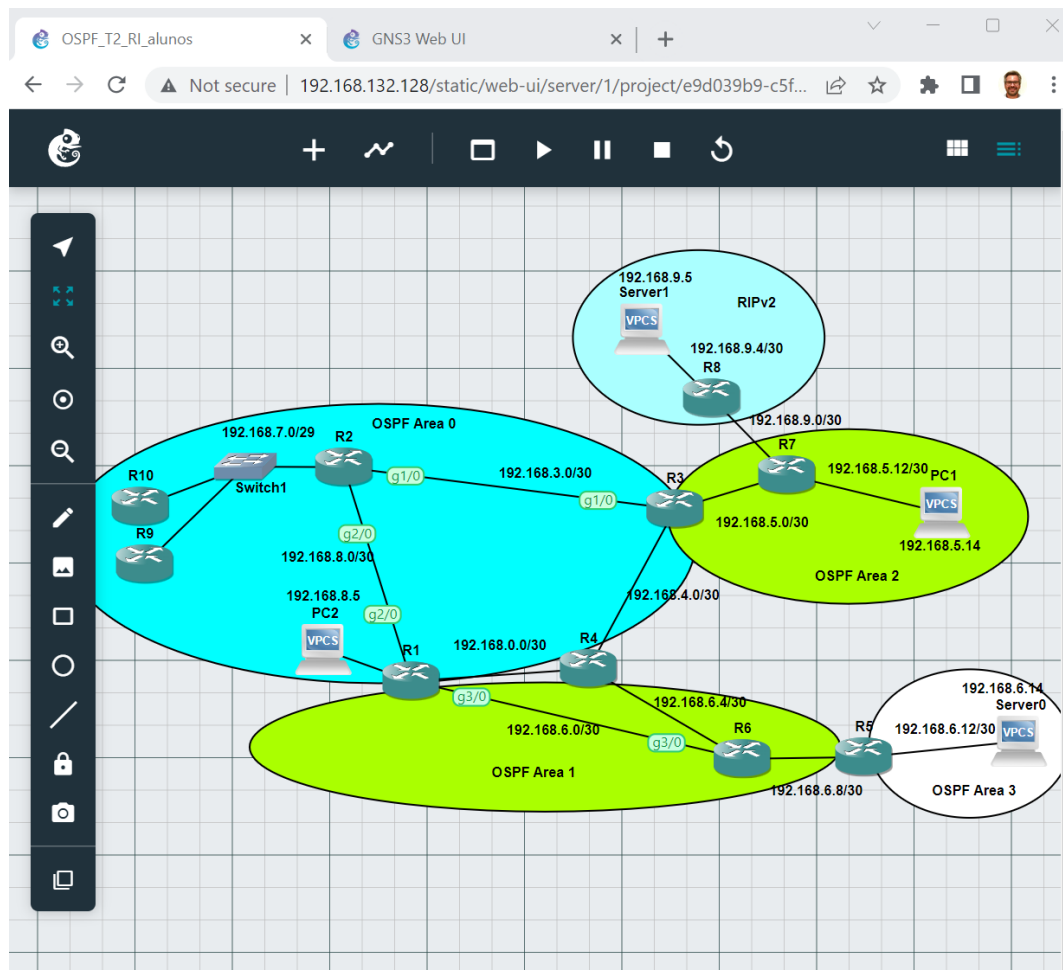


Access to the WEB UI

You can import the support file (portable Project) through the web UI:



Example



VLAN Fundamental Concepts

Definition VLAN: “a single layer-2 network may be partitioned to create multiple distinct broadcast domains, which are mutually isolated so that packets can only pass between them via one or more routers.

Such a domain is referred to as a **Virtual Local Area Network, Virtual LAN or VLAN**”.

VLAN's simplify network management, limit the size of a broadcast domain and improve network efficiency.

VLANs are of two types.

- **Static VLAN** - we assign specific ports to a specific VLAN.
- **Dynamic VLAN** - we assign MAC addresses to a specific VLAN.

In VLAN configuration, there are two types of port configuration: Access Mode and Trunk Mode.

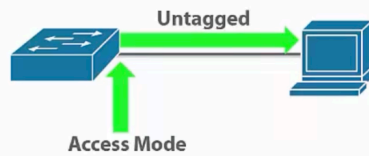
The introduction of tags in VLANs led to a revision of the standard with 802.1Q Encapsulation.

(https://en.wikipedia.org/wiki/IEEE_802.1)

Access Mode Port

(Devices that do not understand VLANs)

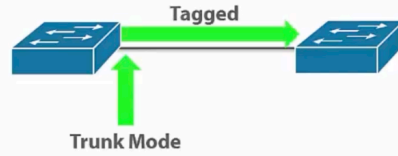
- Computers
- Printers
- Cameras



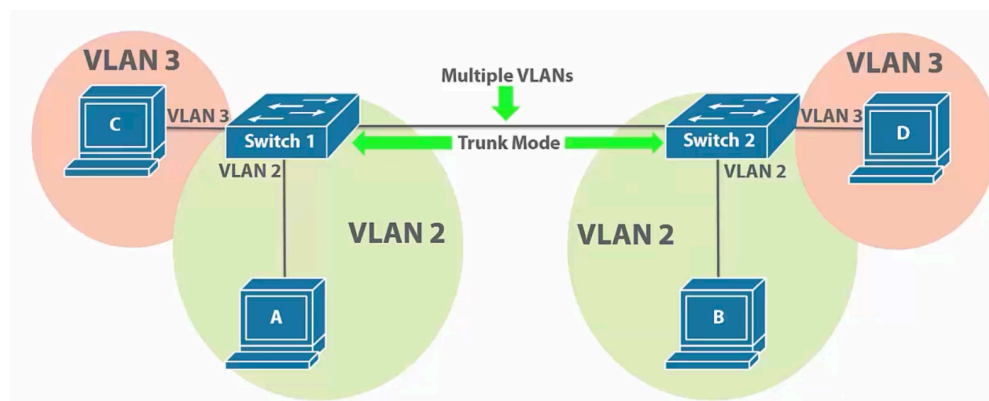
Trunk Mode Port

(Devices that use VLAN tags)

- Switches
- VM Hosts
- Autonomous/FlexConnect APs



Application example:



O padrão IEEE 802.1q traz a necessidade de introduzir campos de identificação nos quadros Ethernet. Essa informação extra será utilizada no roteamento dos quadros, e serão usados pelas pontes ou switches para encaminhar ou filtrar o tráfego recebido. Atendendo a esta necessidade, e à especificação 802.1p (que define uma tag de prioridade dos quadros Ethernet), a norma IEEE 802.3ac, de 1998, acrescentou mais 4 bytes ao tamanho máximo de um quadro Ethernet.

QUADRO ETHERNET SEM A MARCAÇÃO DE VLAN

Preâmbulo	Endereço de destino	Endereço de origem	Tamanho / Tipo	Dados	CRC
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Preâmbulo	Endereço de destino	Endereço de origem	TPID	TCI	Tamanho / Tipo	Dados	CRC
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QUADRO ETHERNET COM A MARCAÇÃO DE VLAN

3 bits	1 bit	12 bits
Prioridade	CFI	VLAN Identificador

The new header fields are:

TPID (Tag Protocol Identifier): 2-byte field which, in the case of a frame corresponding to the 802.1q standard, carries the fixed value 0x8100. This allows this frame to be distinguished from one that does not contain the tag (instead of this field with a fixed value, there is a field corresponding to the size/type of the frame).

TCI (Tag Control Information): 2-byte field subdivided into:

PCP (Priority Code Point): 3 bits referring to the 802.1p standard (Ethernet frame priority), which define a priority level from 0 (lowest) to 7 (highest); by default, the value 1 is used. The priority level attribute is one of the Quality Of Service (QoS) parameters that allows different treatment of data traveling on a network.

Canonical Form Indicator (CFI): 1 bit that indicates canonical form (Ethernet) with a value of "0" and non-canonical form (FDDI, token ring...) with a value of "1". This field is used for compatibility reasons between Ethernet and Token Ring networks. If a frame received on an Ethernet port has CFI 1, then it should not be transmitted as is to an untagged port.

VLAN Identifier (VID): 12 bits indicating the VLAN network to which the frame belongs. The values 0x000 and 0xFFf are reserved; thus, 4094 distinct VLANs can exist.

VLAN - Formato da trama com Tag



- **VLAN Type - Ethernet type 0x8100**

type: indicates higher layer protocol (mostly IP but others possible, e.g., Novell IPX, AppleTalk)

- **Tag Control Info**

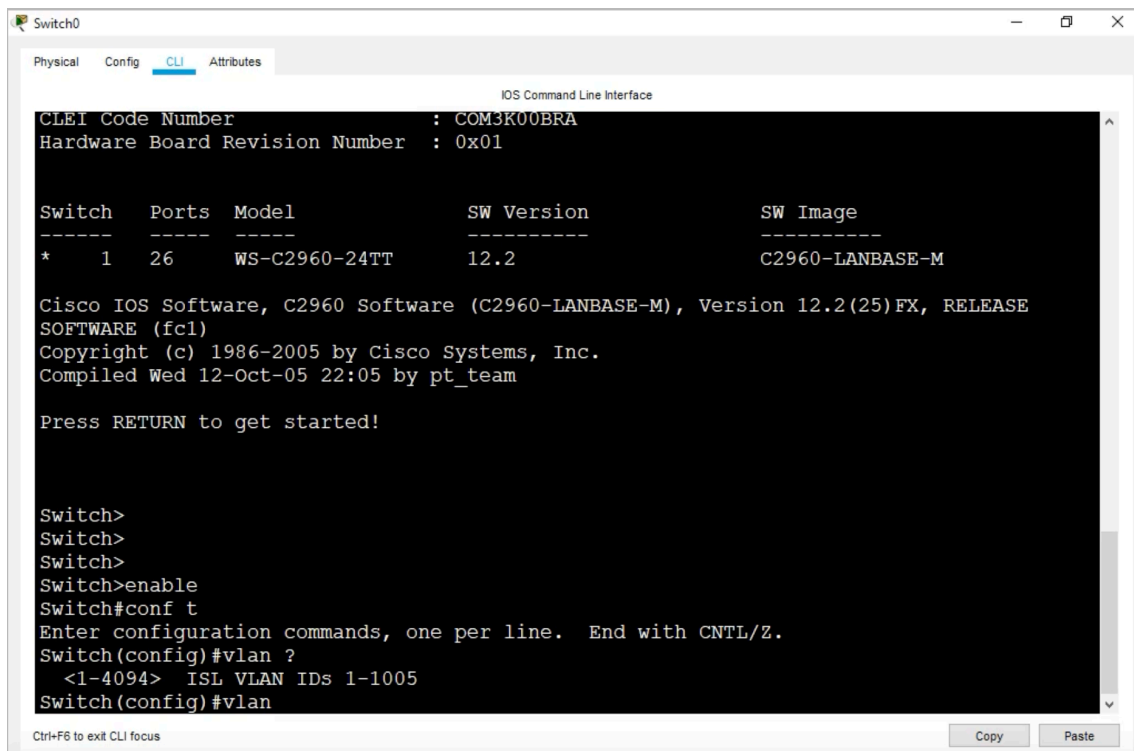
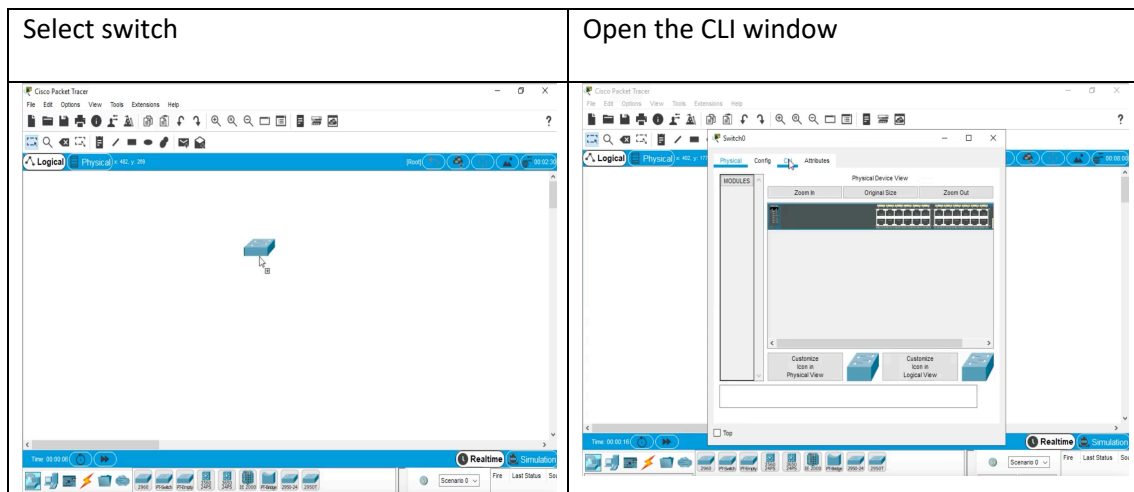
PRI	CFI	VLAN ID
VLAN ID		

- PRI (3bit) - Prioridade 802.1p
- CFI (1bit) - 0 na Ethernet (reservado)
- VLAN ID (12bit) - Identif. da VLAN
 - 0 - sem VID (só prioridade)
 - 1 - ID por omissão
 - 2 até FFE - a atribuir
 - FFF - reservado

- O tamanho máximo da trama passa a **1522 bytes** (excluindo o Preâmbulo+SFD)

Preâmbulo	7 bytes
SFD (start frame delimiter)	1 byte
DA (destination address)	2 ou 6 bytes
SA (source address)	2 ou 6 bytes
VLAN Type	2 bytes
Tag Control Info	2 bytes
Length / Type	2 bytes
Data	46 a 1500 bytes
Pad (opcional)	
FCS (frame check sequence)	4 bytes

Preparations for configuring a VLAN



The command `# vlan ?` allows you to see the possible range for assigning IDs to a VLAN (4094 $\diamond$ 12 bits):

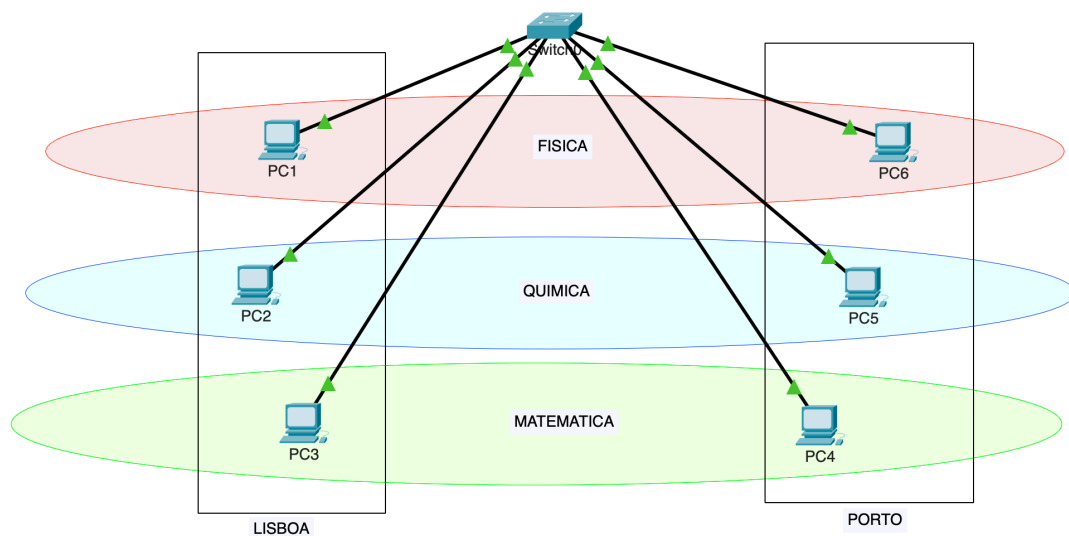
VLAN ID (12bit) - VLAN identifier

- 0 - no VID (priority only)
- 1 - Default ID
- 2 to 4094 (FFE) - to be assigned
- 4095 (FFF) - reserved

In the following exercises, we will create three static VLANs and assign them names and ports. Let's start by creating VLANs for three different departments spread across two physical locations. The following table summarizes the designations and IDs to implement, avoiding VLAN Id=1 as it is reserved for the default VLAN:

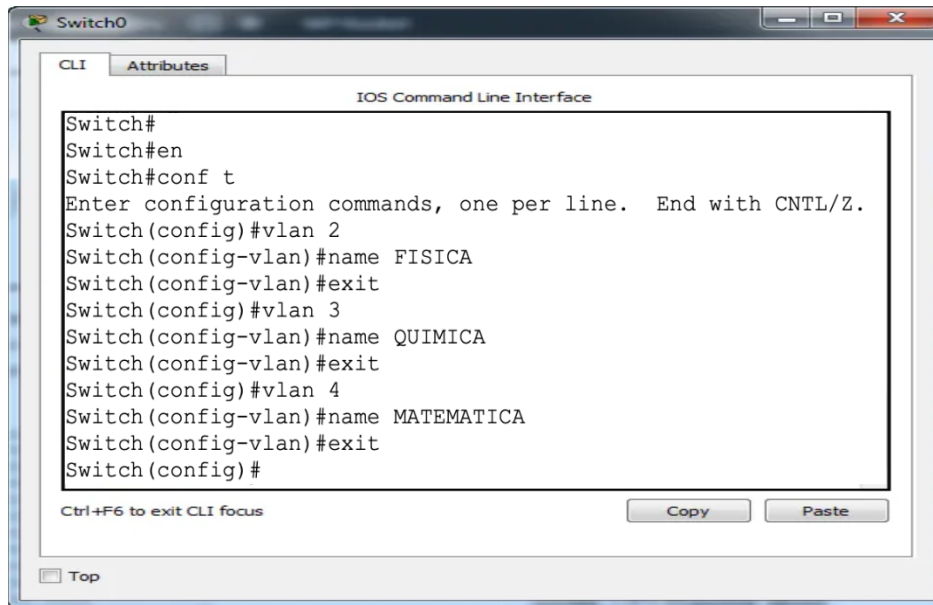
VLAN Id	VLAN Name
2	FISICA
3	QUIMICA
4	MATEMATICA

The intended architecture will be as follows and can be obtained at the following link, where some configurations have already been made.



Configuring the desired VLANs

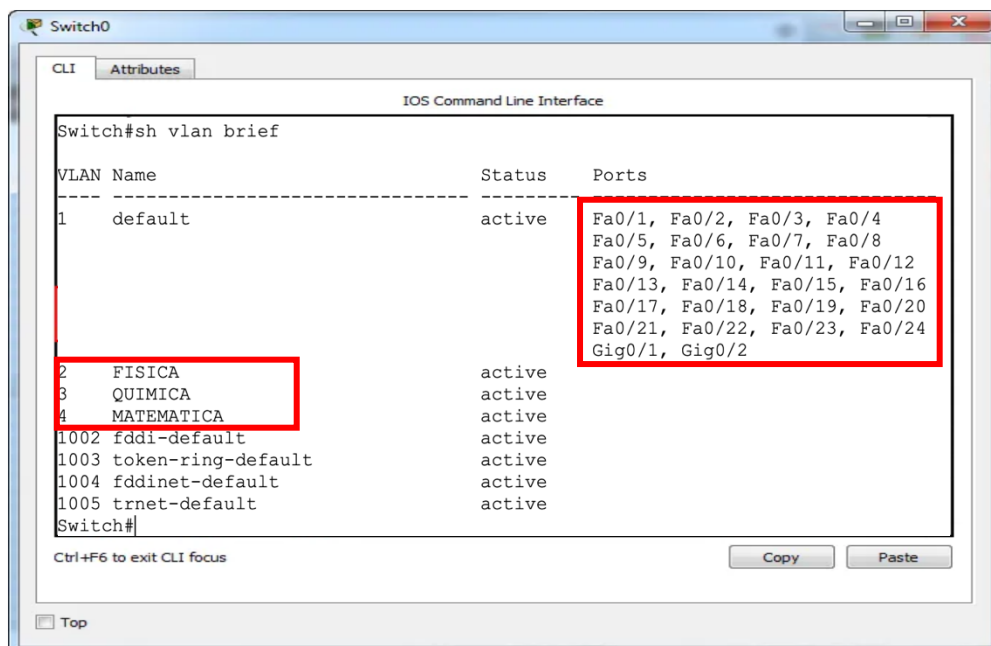
We need to be in the switch0 CLI in global configuration mode to create VLANs:



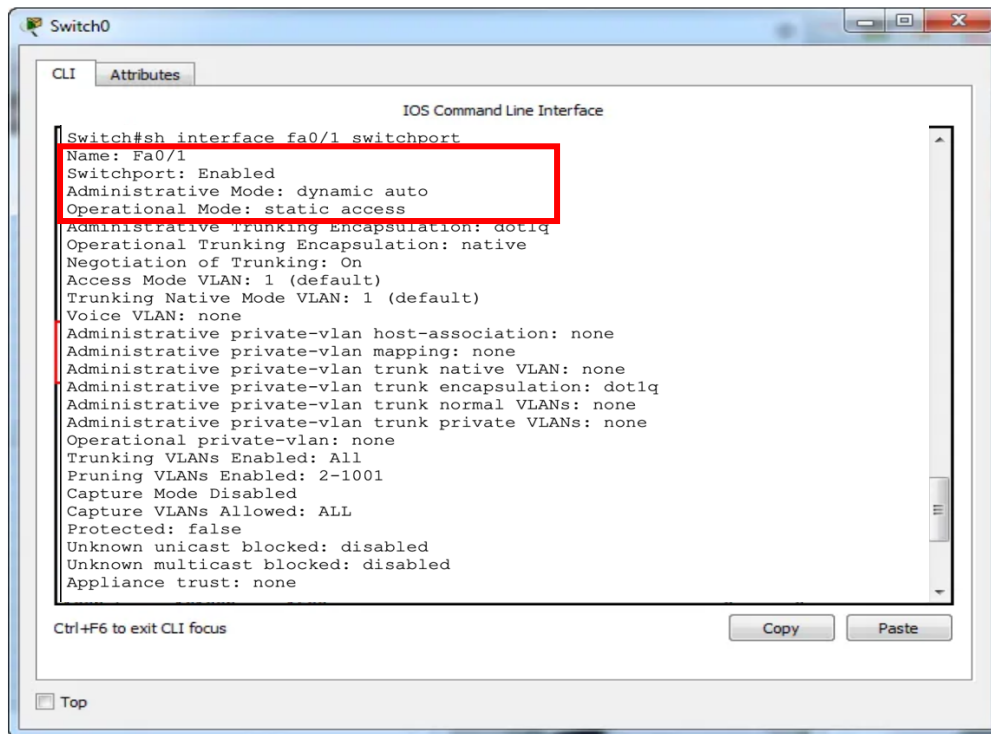
The command “vlan 2” assigns the number 2 to the VLAN we are going to create. (Note that, as we have seen, we can assign any number from 2 to 4094 to the VLAN).

The command “name FISICA” assigns the name “FISICA” to VLAN 2. Similarly, we can create the VLANs QUIMICA and MATEMATICA.

Finally, we can use the command “show vlan” or “show vlan brief” to view the settings we have made. We can also query a specific VLAN, e.g., “show vlan id x.”



We can also use the command, e.g., “show interfaces fa0/1 switchport,” which produces a report at the interface configuration level.



Assigning ports to the VLAN

Before assigning ports to the VLAN, let's look at the port statuses on the switch. As we saw in the previous point, all switch ports are associated with VLAN 1 by default. This is a special VLAN used for administration. It is the only pre-created VLAN on the Cisco switch. All ports are members of this VLAN by default. Therefore, when you assign a port to a specific VLAN, you are simply changing the port from VLAN 1 to VLAN X.

Now let's assign ports to the VLANs we just created. First, let's assign three ports to the "FISICA" VLAN using the "interface int0x" command. Alternatively, you can combine this instruction with the range command, resulting in "interface range int01-int0X."

Let's add fast ethernet ports 1 to 6 to their respective VLANs. Example:


```
Switch(config-if)#
Switch(config-if)#int fastEthernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#int fastEthernet 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#int fastEthernet 0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
```

Ctrl+F6 to exit CLI focus

Copy Paste

Top

Now let's check the port statuses. The ports should already be associated with their respective VLANs according to the implemented network topology.

Switch0

CLI Attributes

IOS Command Line Interface

```
Switch#sh vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
2 FISICA	active	Fa0/1, Fa0/6
3 QUIMICA	active	Fa0/2, Fa0/5
4 MATEMATICA	active	Fa0/3, Fa0/4
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Switch#

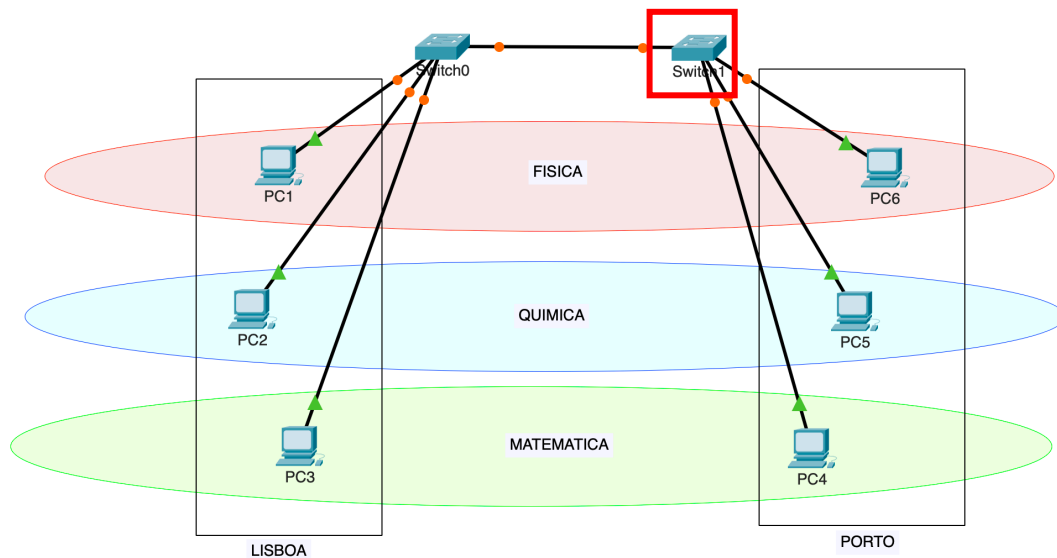
Ctrl+F6 to exit CLI focus

Copy Paste

Top

Configuring ports in trunk mode

Now modify the initial configuration by adding switch1 and configuring the intermediate ports in trunk mode to enable tagging of traffic from multiple VLANs. This new configuration can be obtained at link, where an additional switch has been added.

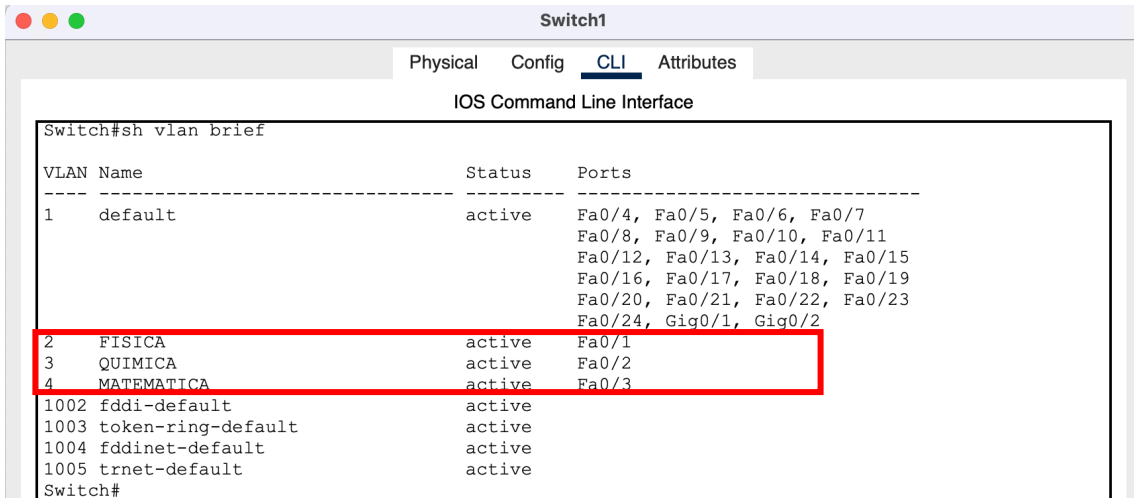


Let's start by adjusting the necessary settings on switch 0 and configuring switch 1:

Desired settings on switch 0:

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch#sh vlan brief
VLAN Name      Status Ports
-----
1  default      active Fa0/4, Fa0/5, Fa0/6, Fa0/7
                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                Fa0/24, Gig0/1, Gig0/2
2  FISICA       active Fa0/1
3  QUIMICA      active Fa0/2
4  MATEMATICA   active Fa0/3
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
```

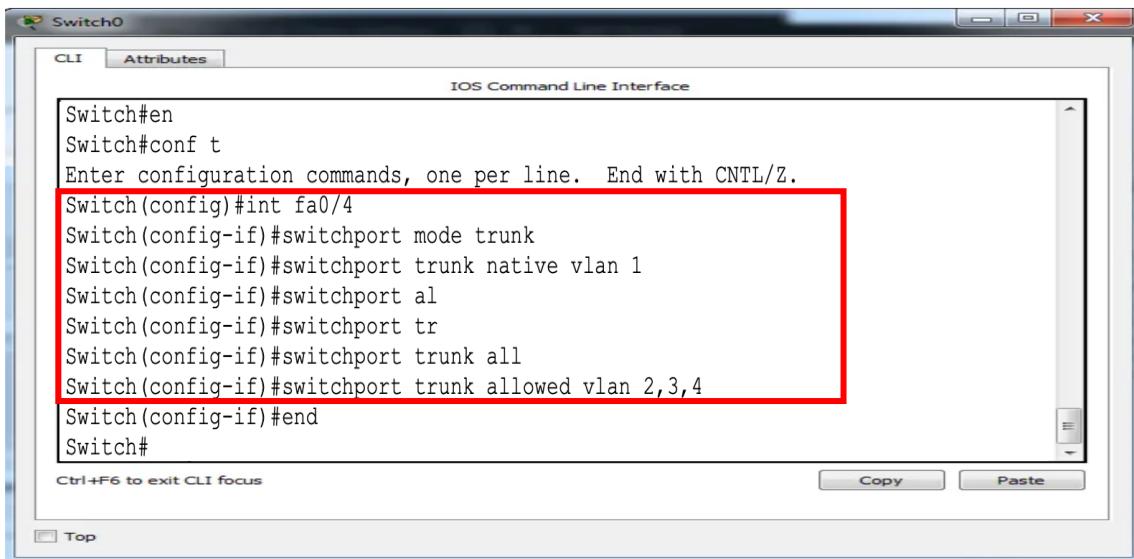
Similarly, the settings on switch 1 should be as follows:



The screenshot shows the CLI of Switch1 with the 'CLI' tab selected. The command 'Switch#sh vlan brief' has been executed, displaying a table of VLANs. A red box highlights the first four VLANs: VLAN 2 (FISICA), VLAN 3 (QUIMICA), and VLAN 4 (MATEMATICA), all of which are active and assigned to specific ports.

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/1, Gig0/2
2	FISICA	active	Fa0/1
3	QUIMICA	active	Fa0/2
4	MATEMATICA	active	Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

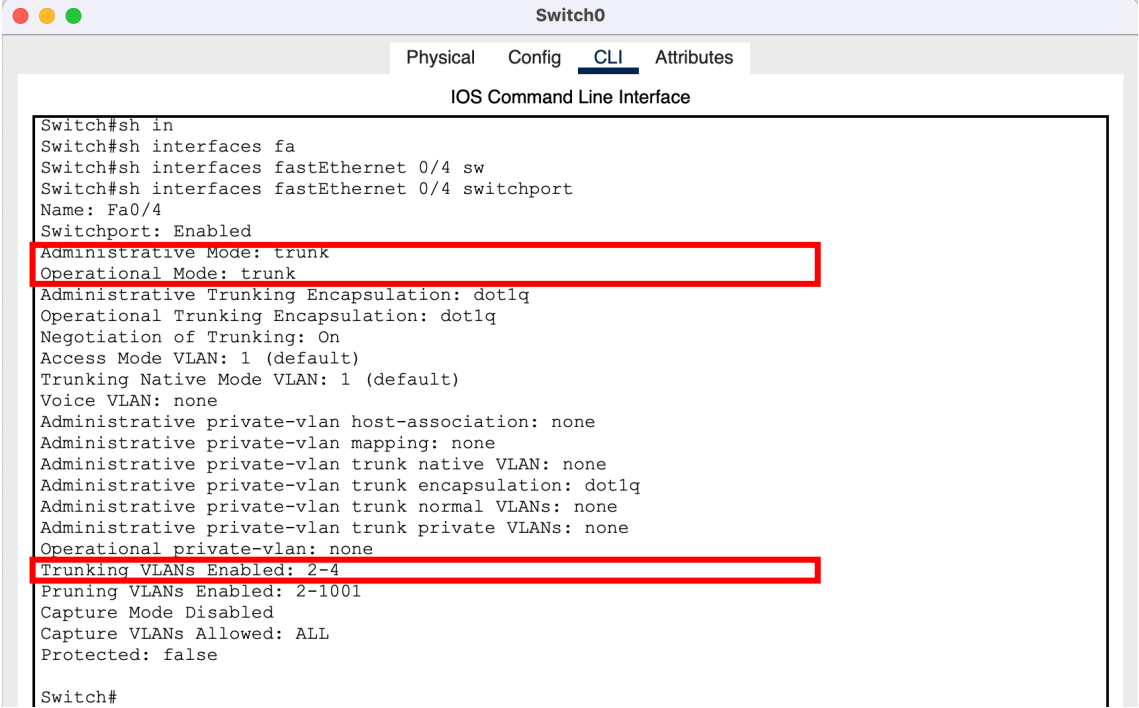
Next, the configuration of TRUNK mode on switch 0 is illustrated. Similar configurations will be necessary on switch 1.



The screenshot shows the CLI of Switch0 with the 'CLI' tab selected. The user has entered several commands to configure interface fa0/4 as a trunk port. A red box highlights the configuration commands: 'int fa0/4', 'switchport mode trunk', 'switchport trunk native vlan 1', 'switchport al', 'switchport tr', 'switchport trunk all', and 'switchport trunk allowed vlan 2,3,4'.

```
Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 1
Switch(config-if)#switchport al
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk all
Switch(config-if)#switchport trunk allowed vlan 2,3,4
Switch(config-if)#end
Switch#
```

Next, the configuration must be validated using the command “show interfaces fa0/4 switch port”:

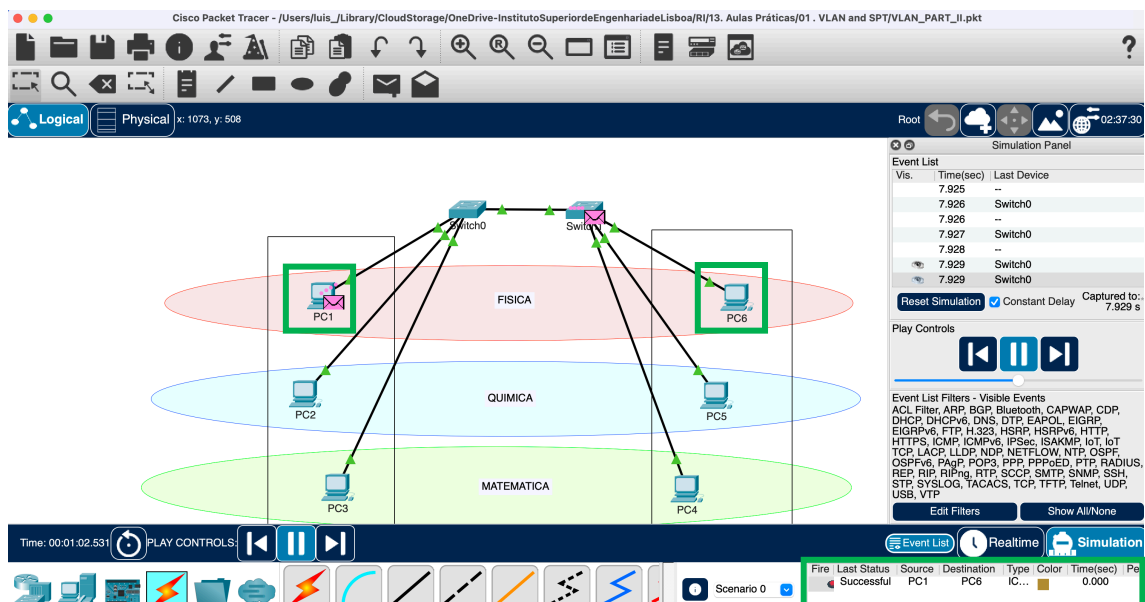


```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

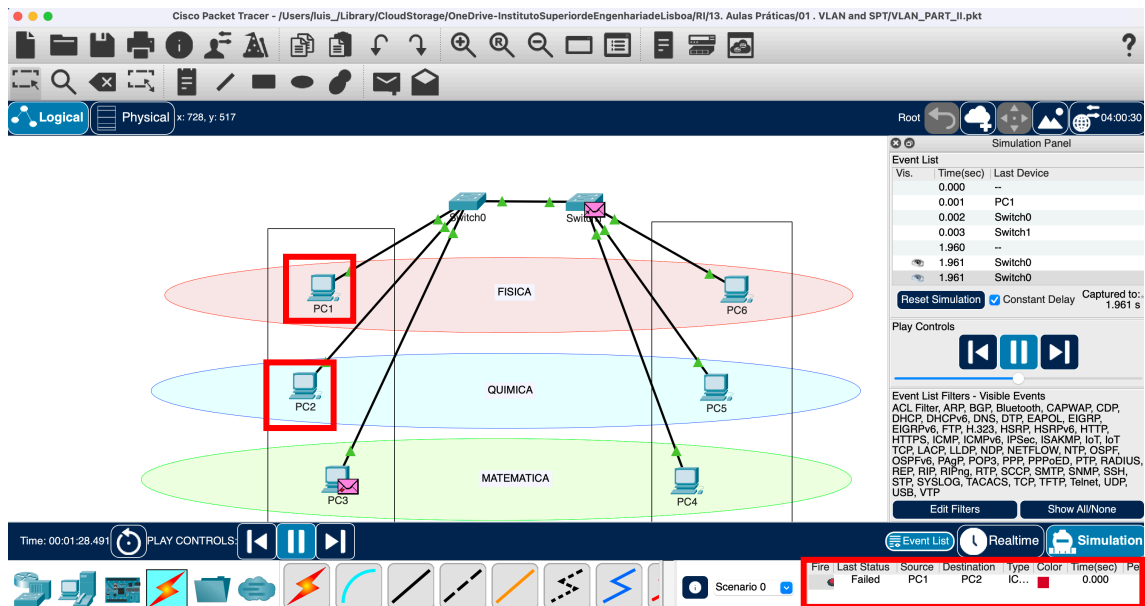
Switch#sh in
Switch#sh interfaces fa
Switch#sh interfaces fastEthernet 0/4 sw
Switch#sh interfaces fastEthernet 0/4 switchport
Name: Fa0/4
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: 2-4
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false

Switch#
```

Verify that there is connectivity between PC1 and PC6 (both on the FISICA VLAN):



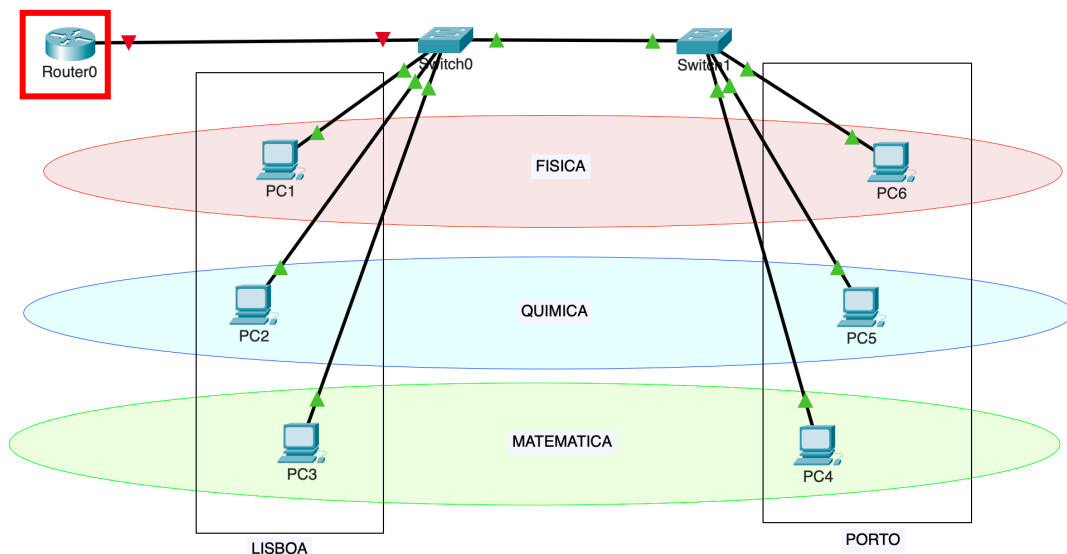
Now verify that there is no connectivity between PC1 and PC2 (they belong to different VLANs). Remember that there is no router in the topology to ensure packet forwarding between different VLANs!



Implementing a “Router on a Stick” configuration

(https://en.wikipedia.org/wiki/Router_on_a_stick)

Next, a router should be added to the previous configuration in order to enable packet forwarding between different VLANs. The router on a stick (ROAS) configuration, also known as a one-armed router, refers to a router with a single physical or logical connection to a network, as shown in the following figure. This topology can be obtained in link, where the router has been added.



To implement the desired topology, configure the subinterfaces of each VLAN on the router, as shown below:

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g
Router(config)#int gigabitEthernet 0/0/1
Router(config-if)#description TRUNK LINK
Router(config-if)#exit
Router(config)#int g0/0/1.2
Router(config-subif)#enc
Router(config-subif)#encapsulation d
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.2.100 255.255.255.0
Router(config-subif)#int g0/0/1.3
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.100 255.255.255.0
Router(config-subif)#int g0/0/1.4
Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.4.100 255.255.255.0
Router(config-subif)#

```

Check the settings on the router interfaces using the sh ip interfaces brief command.

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router#sh ip int
Router#sh ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 unassigned YES unset administratively down down
GigabitEthernet0/0/1 unassigned YES unset administratively down down
GigabitEthernet0/0/1.2 192.168.2.100 YES manual administratively down down
GigabitEthernet0/0/1.3 192.168.3.100 YES manual administratively down down
GigabitEthernet0/0/1.4 192.168.4.100 YES manual administratively down down
GigabitEthernet0/0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#

```

Next, the Ge 0/1 interface must be configured for TRUNK mode on switch 0:

```

Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int g
Switch(config)#int gigabitEthernet 0/1
Switch(config-if)#sw
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Switch(config-if)#

```

You can complement this with the `# switchport nonegotiate` command to force the protocol used to always be 802.1Q, rather than ISL. Confirm with `# sh interfaces trunk`.

The final step will be to assign IP addresses to the PCs and define the gateway, as previously defined. Note: Don't forget to shut down the interfaces.

Check again for connectivity between PC1 and PC5 (different VLANs) and confirm that with the "router on a stick" configuration, there is now connectivity between different VLAs (see simulation mode in PT:

